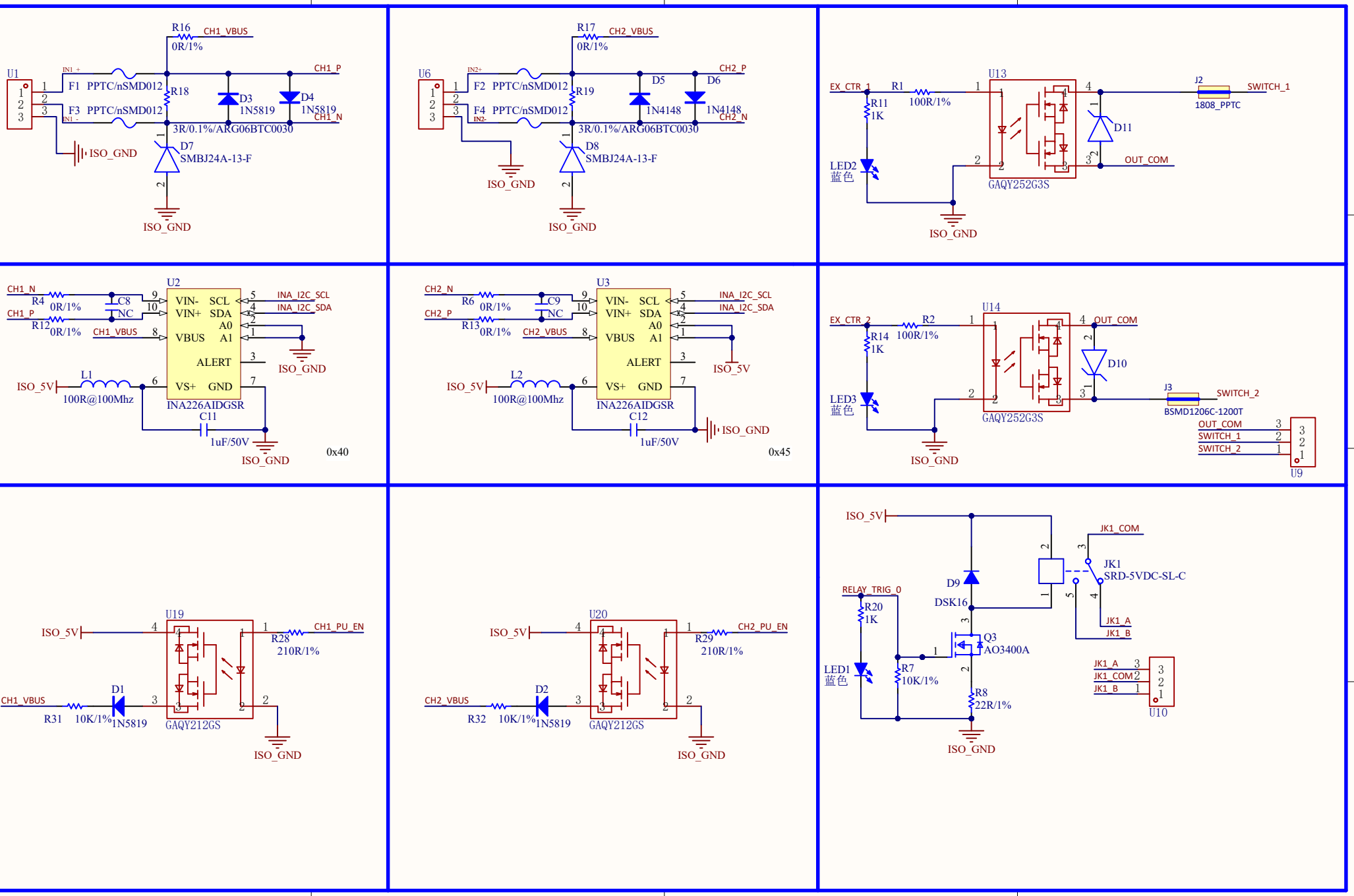
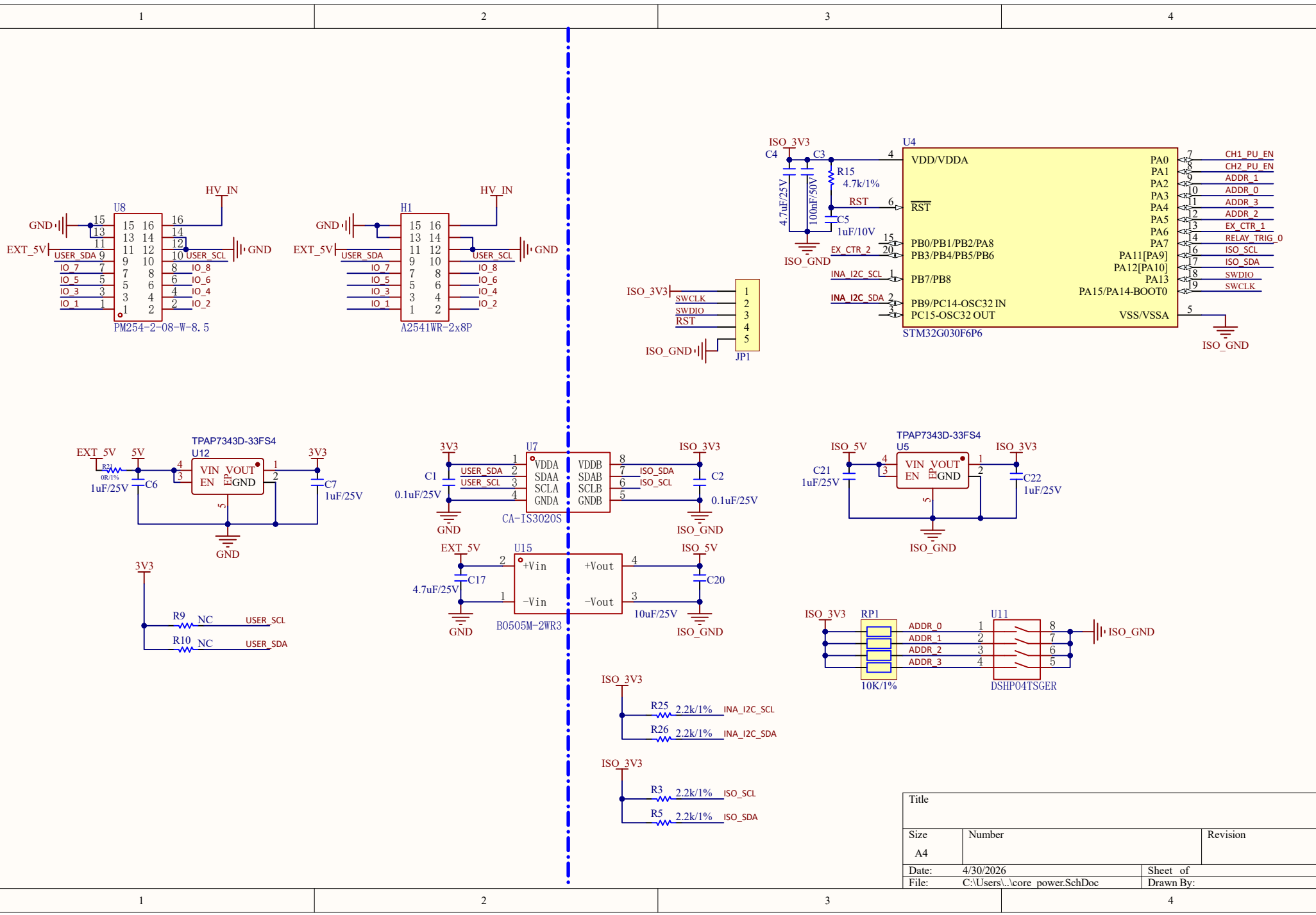

U1
R16 0R/1% CH1_VBUS
F1 PPTC/nSMD012
F3 PPTC/nSMD012
R18
D3 1N5819
D4 1N5819
CH1_P
CH1_N
3R/0.1%/ARG06BTC0030
D7 SMBJ24A-13-F
ISO_GND
U6
R17 0R/1% CH2_VBUS
F2 PPTC/nSMD012
F4 PPTC/nSMD012
R19
D5 1N4148
D6 1N4148
CH2_P
CH2_N
3R/0.1%/ARG06BTC0030
D8 SMBJ24A-13-F
U13
EX_CTR_1
R1 100R/1%
R11 1K
LED2 蓝色
GAQY252G3S
D11
J2 1808_PPTC
SWITCH_1
OUT_COM
U2
CH1_N
CH1_P
R4 0R/1%
R12 0R/1%
C8 NC
CH1_VBUS
VIN- SCL
VIN+ SDA
A0
VBUS A1
ALERT
VS+ GND
INA_I2C_SCL
INA_I2C_SDA
L1 100R@100Mhz
ISO_5V
INA226AIDGSR
C11 1uF/50V
0x40
U3
CH2_N
CH2_P
R6 0R/1%
R13 0R/1%
C9 NC
CH2_VBUS
L2 100R@100Mhz
C12 1uF/50V
0x45
U14
EX_CTR_2
R2 100R/1%
R14 1K
LED3 蓝色
D10
J3 BSMD1206C-1200T
SWITCH_2
OUT_COM
SWITCH_1
U9
U19
R28 210R/1%
CH1_PU_EN
D1 1N5819
R31 10K/1%
GAQY212GS
U20
R29 210R/1%
CH2_PU_EN
D2 1N5819
R32 10K/1%
D9 DSK16
RELAY_TRIG_0
R20 1K
LED1 蓝色
R7 10K/1%
Q3 AO3400A
R8 22R/1%
JK1 SRD-5VDC-SL-C
JK1_COM
JK1_A
JK1_B
U10


U8
HV_IN
GND
EXT_5V
USER_SDA
USER_SCL
IO_7
IO_5
IO_3
IO_1
IO_8
IO_6
IO_4
IO_2
PM254-2-08-W-8.5
H1
A2541WR-2x8P
ISO_3V3
SWCLK
SWDIO
RST
ISO_GND
JP1
U4
VDD/VDDA
PB0/PB1/PB2/PA8
PB3/PB4/PB5/PB6
PB7/PB8
PB9/PC14-OSC32 IN
PC15-OSC32 OUT
PA0
PA1
PA2
PA3
PA4
PA5
PA6
PA7
PA11[PA9]
PA12[PA10]
PA13
PA15/PA14-BOOT0
VSS/VSSA
CH1_PU_EN
CH2_PU_EN
ADDR_1
ADDR_0
ADDR_3
ADDR_2
EX_CTR_1
RELAY_TRIG_0
ISO_SCL
ISO_SDA
SWDIO
SWCLK
EX_CTR_2
INA_I2C_SCL
INA_I2C_SDA
R15
4.7k/1%
C4
C3
C5
1uF/10V
4.7uF/25V
100nF/50V
STM32G030F6P6
TPAP7343D-33FS4
U12
5V
3V3
C6
C7
1uF/25V
R2
0R/1%
U7
CA-IS3020S
VDDA
SDAA
SCLA
GNDA
VDDB
SDAB
SCLB
GNDB
C1
C2
0.1uF/25V
U15
B0505M-2WR3
+Vin
-Vin
+Vout
-Vout
C17
C20
4.7uF/25V
10uF/25V
R9
R10
NC
U5
C21
C22
RP1
10K/1%
U11
DSHP04TSGER
R25
R26
R3
R5
2.2k/1%
Title
Size
A4
Number
Revision
Date:
4/30/2026
Sheet of
File:
C:\Users\...\core_power.SchDoc
Drawn By: